Reg. No : 2116241001017

Question Paper Code :051071

B.E./B.Tech DEGREE EXAMINATION APRIL/MAY - 2025

Second Semester

Computer Science and Engineering

CS23231 DATA STRUCTURES

(Common to: ECE, EEE, IT, AI&ML, CSD, AI&DS, CSE(CS))

(Regulations 2023)

Time : Three Hours | Maximum : 100 Marks

Answer ALL Questions

PART A (10x2=20 Marks)

1.	Why data structures are important in computer science?	[CO1]	[A1]
2.	Define double circularly linked list.	[CO1]	[A1]
3.	Translate the expression A+B-C+D to postfix form.	[CO1]	[A2]
4.	Develop a routine for insertion operation of singly linked list.	[CO1]	[B1]
5.	List the steps involved in deleting a node which has 1 children from a binary search tree.	[CO2]	[A1]
6.	List the applications of non-linear data structures.	[CO2]	[A1]
7.	When a graph is said to be strongly connected?	[CO2]	[A2]
8.	Compare directed graph and undirected graph.	[CO2]	[A2]
9.	Distinguish between merge sort and quick sort.	[CO3]	[A2]
10.	Define hash function. Classify the different types of hash function.	[CO3]	[A1]

PART B (5x13= 65 Marks)

11. a. Explain the insertion operation in singly linked list. How nodes are inserted after a specified node? [CO1] [B1]

[OR]

b. Develop a routine to insert and delete an element at the beginning in doubly linked list with suitable example. [CO1] [B1]

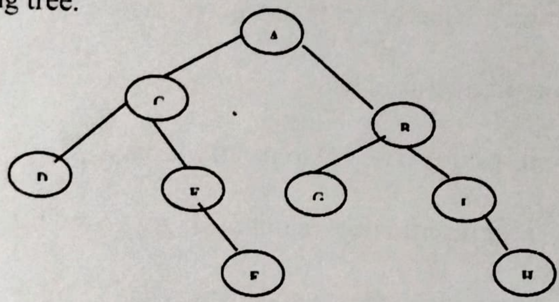
12. a. Examine the various operations are performed of stack with examples. [CO1] [B2]

[OR]

b. List and explain the various operations performed on queue. Write a routine to implement enqueue and dequeue operations. [CO1] [B2]

13. a.i). Apply the different types of tree traversal techniques using the following tree. [CO2] [B1]

(6)



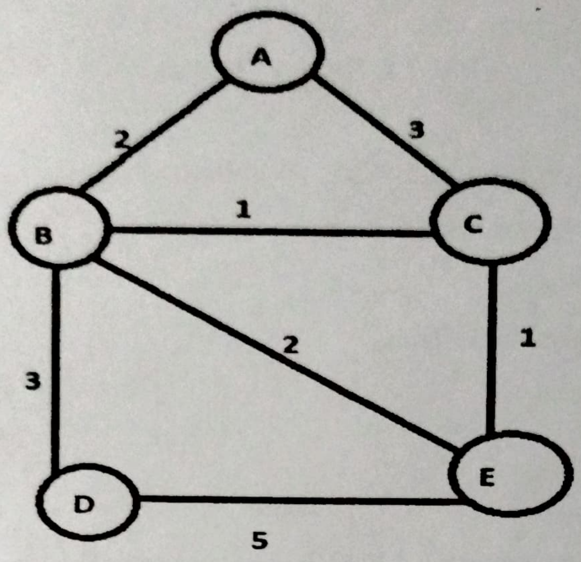
ii). Construct an expression tree for the expression (a + b * c) +((d * e + 1) * g). Give the outputs when you apply preorder, in order and post order traversals. [CO2] [B1]

(7)

[OR]

b. Construct a binary search tree for the following numbers start from an empty binary search tree. 45, 26, 10, 60, 70, 30, 40 Delete keys 10, 60 and 45 one after the other and show the trees at each stage. [CO2] [B1]

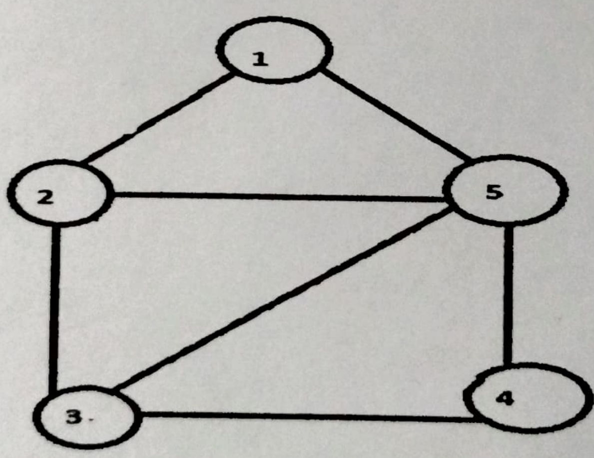
14 a Illustrate with a neat graph explain the prims algorithm for finding the minimum spanning tree. [CO2] [B2]



[OR]

b.i. Compare breadth first search with depth first search. (6) [CO2] [B2]

ii. Write and trace the BFS algorithms with the following Graph and explain the routine for implementing it. (7) [CO2] [B2]



15 a. Analyze the algorithm of insertion sort by sorting the following set of numbers as an example: 17 26 54 77 93 31 44 55 30. [CO4] [B2]

[OR]

b. Examine the various collision handling techniques with an example. [CO3] [B2]

PART C (1x15=15 Marks)

16. a. Develop a C program to perform the following operations in a Circular linked list with a neat sketch. [C1]

A. Insert at the end.

B. Insert at specified location.

C. Delete a node.

D. Display.

[OR]

b. Translate the expression A * B + (C – D / E) to equivalent postfix notation. Write the program to implement this. [C1]

Reg. No : 2 1 1 6 2 3 1 0 0 1 2 5 5

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Computer Science and Engineering

CS23231 DATA STRUCTURES

(Common to Electrical and Electronics Engineering, Electronics and Communication Engineering, Information Technology, Computer science and Design, Cyber security, Artificial Intelligence and Machine learning, Artificial Intelligence and Data Science)

(Regulations 2023)

Time: Three Hours Maximum: 100 Marks

Answer ALL Questions

PART A (10x2=20 Marks)

1. Define self-referential structure with an example. [CO1] [A1]
2. Define ADT. [CO1] [A2]
3. Compare and contrast stack and queue. [CO2] [A1]
4. Define circular queue. [CO2] [A2]
5. How many nods are there in full binary tree if the height of tree is 3? [CO3] [A1]
6. What is binary search tree. Give an example, [CO3] [A2]
7. Compare tree and graph data structures. [CO4] [A2]
8. Define degree in a graph. [CO4] [A1]
9. Define mid square method in collision resolution techniques. [CO5] [A2]
10. Define hashing and state its importance in data structures. [CO5] [A1]

PART B (5x13= 65 Marks)

11.	a.	Write a C program to dynamically allocate memory for an array of integers, and demonstrate insertion and deletion operations on this array.	[CO1]	[B2]
		[OR]		
	b.	Given a set of integers, use a circular linked list to store and sort them. Write a C program that implements the circular linked list and sorts the numbers in ascending order.	[CO1]	[B2]
12.	a.	Implement a stack using an array in C. Use this stack to evaluate the postfix expression 6 3 2 4 + - *. Show the step-by-step evaluation process.	[CO2]	[B1]
		[OR]		
	b.	You have a sequence of customer IDs that need to be processed in the order they arrived. Use a linked list to implement a queue that manages this sequence. Write a C program to enqueue, dequeue, and display the customer IDs.	[CO2]	[B2]
13.	a.	Explain tree traversal techniques with diagrams and appropriate routines.	[CO3]	[B1]
		[OR]		
	b.	You are given a set of operations to be performed on an AVL tree: Insert the following elements in the given order: 30, 20, 40, 10, 25, 35, 50. Write the steps to insert each element and show the rotations required to maintain the AVL property, with appropriate routines.	[CO3]	[B2]
14.	a.	Consider the following graph represented by an adjacency matrix. Apply Depth First Search (DFS) to visit all the nodes starting from node A. Show the step-by-step traversal order. Implement this in C.	[CO5]	[B1]

$$
\begin{array}{ccccc}
 & A & B & C & D \\
A & 0 & 1 & 1 & 0 \\
B & 1 & 0 & 0 & 1 \\
C & 1 & 0 & 0 & 1 \\
D & 0 & 1 & 1 & 0
\end{array}
$$

[OR]

b. You have a weighted graph representing a network of cities connected by roads. Apply Prim's algorithm to find the Minimum Spanning Tree starting from node 0. Given the adjacency matrix below, show the step-by-step process and implement the solution in C. [CO5] [B2]

	A	B	C	D
A	0	1	2	0
B	1	0	3	4
C	2	3	0	5
D	0	4	5	0

15. a. You are given an unsorted list of integers: [15, 8, 24, 2, 19, 30]. Apply Merge Sort to sort the list and explain the division and merging process in each step. Implement this in C and print the sorted list. [C1] [B2]

[OR]

b. Given input (4371.1323,6173,4199,4344,9679,1989) and a hash function $h(x) \equiv x(mod\ 10)$, Show the resulting [C2] [B2]

(i) Open hash table. (4)

(ii) Closed hash table using linear probing. (5)

(iii) Close hash table using quadratic probing. (4)

PART C (1x15=15 Marks)

16.a Given two sorted linked list namely L_1 and L_2, compute $L_1 \cup L_2$ and $L_1 \cap L_2$. [C1]

[OR]

b. Explain deleting an element from BST for the following with neat diagram and routine. [C1]

(i) node with no children.

(ii) node with one child.

(iii) node with two children.

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B.E./B.Tech DEGREE EXAMINATION NOV/DEC - 2024

Second Semester

Computer Science and Engineering

CS23231 DATA STRUCTURES

(Common to Electrical and Electronics Engineering, Electronics and Communication Engineering, Information Technology, Computer science and Design, Cyber security, Artificial Intelligence and Machine learning, Artificial Intelligence and Data Science)

(Regulations 2023)

Time: Three Hours　　　　Maximum: 100 Marks

Answer ALL Questions

PART A (10x2=20 Marks)

1. Name two applications of singly linked lists. [CO1] [A1]
2. Define ADT. [CO1] [A2]
3. What is the result of enqueue operation? [CO2] [A1]
4. Name the operation used to add an element to a stack. [CO2] [A2]
5. Define Full binary tree. [CO3] [A1]
6. What is binary search tree. Give an example, [CO3] [A2]
7. Compare tree and graph data structures. [CO4] [A2]
8. State the main purpose of Dijikstra's algorithm. [CO4] [A1]
9. Define mid square method in collision resolution techniques. [CO5] [A2]
10. Define the term "load factor" in the context of hashing. [CO5] [A1]

PART B (5x13= 65 Marks)

11. a. Demonstrate the process of dynamic memory allocation with a code snippet. [CO1] [B2]

[OR]

b. Given a set of integers, use a circular linked list to store and sort them. Write a C program that implements the circular linked list and sorts the numbers in ascending order. [CO1] [B2]

12. a. Elaborate on the circular queue concept and its implementation in an array with examples. [CO2] [B1]

[OR]

b. Discuss stack and its implementation. [CO2] [B2]

13. a. Explain tree traversal techniques with diagrams and appropriate routines. [CO3] [B1]

[OR]

b. Describe the algorithm for inserting an element in a binary search tree with neat diagram. [CO3] [B2]

14. a. Apply Prim's algorithm to find the minimum spanning tree for the following graph. Show each step and the final MST. [CO4] [B1]

M - 1 - N
M - 4 - O
N - 2 - P
O - 3 - P
O - 6 - Q

[OR]

b. You are given a directed acyclic graph (DAG) representing tasks and dependencies. Perform a topological sort on the graph below: [CO4] [B2]

A -> B
A -> C
B -> D
C -> D

Explain each step.

15. a. You are given an unsorted list of integers: [10,18, 24, 2, 11, 3]. Apply Merge Sort to sort the list and explain the division and merging process in each step. Implement this in C and print the sorted list. [C05] [I

[OR]

b. Resolve collisions using open addressing with quadratic probing for keys [12, 22, 32, 42] in a hash table of size 7 using the hash function h(x) = x % 7. Show each step. [C05] [B2]

PART C (1x15=15 Marks)

16.a Illustrate the open addressing technique in hash tables, covering linear probing, quadratic probing, and double hashing with examples. [C1]

[OR]

b. Explain deleting an element from BST for the following with neat diagram and routine. [C1]

(i) node with no children.

(ii) node with one child.

(iii) node with two children.